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LAB PROGRAM - 4

Implement circular queue using array of integers. Provide the following operations:
Insert
Delete
Display
Program should print appropriate messages for empty queue, full queue and queue full conditions.

Pseudo code :

```
set front = rear = -1;

Enque

If (front = -1 && rear = -1)
    set front = rear = 0;
    queue[rear] = x;

else if (rear+1 % N == front)
    printf("queue overflow");

else {
    rear = rear+1 % N;
    queue[rear] = x;

Deque

If (front == -1 && rear == -1)
    printf("queue underflow");

else if( front == rear)
    printf("deque = %d", queue[front]);
    front = front+1 % N;
```

else {

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```c
else{
    printf("deque = %d", queue[front]);
    front = front+1%N;
}
```

display

```c
for(int i = front; i<=rear; (i+1)%N)
    printf("queue is %d", queue[i]);
```

8/ 3/11/23

Code :

```c
#include <stdio.h>
#include <ctype.h>

#define n 5
int queue[5];
int front = -1;
int rear = -1;

void enque(int x){

    if (front == -1 && rear == -1) {
        front = rear = 0;
        queue[rear] = x;
    }
    else if ((rear+1)%N == front){
        printf("queue overflow\n");
    }
    else{
        rear = (rear+1)%N;
        queue[rear] = x;
    }
}
```

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```c
void deque(){
    if (front == -1 && rear == -1){
        printf("queue underflow\n");
    }
    else if(front == rear){
        printf("deque = %d \n", queue[front]);
        front = rear = -1;
    }
    else{
        printf("deque = %d\n", queue[front]);
        front = front+1 % N;
    }
}

void display(){

    for(int i=front; ; i = (i+1) % N){
        printf("%d\t", queue[i]);
        if (i == rear)
        break;
    }
    printf("\n");
}

int main(){
    int ch;
    printf("1:enque \n 2. deque \n 3. display \n");

    while(1){
        int x;
        printf("enter choice");
        scanf("%d", &ch);
```

execute
3/11/25

```
switch(ch){
    case 1: printf("enter number to insert: ");
            scanf("%d", &x);
            enque(x);
            break;
    case 2: deque();
            break;
    case 3: display();
            break;
    case 4: return 0;
    default: printf("Invalid choice");
  }
}
 return 0;
}
```

output :

1. enque
2. deque
3. display

executed 3/11/25

enter choice : 2
queue underflow
enter choice : 1
enter number to insert : 23
enter choice : 1
enter number to insert : 45
enter choice : 1
enter number to insert : 65
enter choice : 1
enter number to insert : 68
enter choice : 1
enter number to insert : 99
enter choice : 1
enter number to insert : 33

queue overflow
enter choice : 2
deque = 23
enter choice : 2
deque = 45
enter choice : 3
65 68 99
enter choice : 4

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